

Samuel **Lucas**

Great at Python, afraid of snakes.

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Summary

Highly motivated engineer with 14+ years of professional programming experience and 10+ years of experience in designing scalable, cloud-based solutions. Expertise in applied AI, team leadership, and building secure, compliant software for safety-critical systems. Proven track record of driving innovation, managing multidisciplinary teams, and successfully leading end-to-end product development from vision to deployment.

Work Experience

Capital One

Lead Software Engineer (Jan 2024 – Present)
- Individual contributor under the Associate Experience line of business, primarily dedicated to compliance platforms
- Designed and implemented several critical business controls, and a new AI-powered platform for SOX testing
- Leader of AI Community of Practice for our line of business, which helps define standard/best practices
- Leader of planning and execution for several recurring hackathons and AI innovation events

Promenade Software

Chief Technology Officer, CypherMed Cloud LLC (Jan 2024 – Dec 2024)
- Worked with the President of Promenade Software to establish a new subsidiary LLC focused on our SaaS product
- Leading development of our SOC2 Type 2 compliant API, multiple SDKs, administrative dashboard, and landing page

Lead Software Engineer (June 2021 – Dec 2024)
- Worked with clients to design and develop highly secure, HIPAA-compliant software for medical devices
- Manage highly scalable, tightly controlled, and autonomously deployed microservice infrastructure built on AWS
- Overseeing design efforts for most of our client projects, monitoring developer operations & mentored junior devs
- Projects range from desktop and mobile apps, to web apps, to complex cloud-based networks of microservices

Product Owner, CypherMed Cloud (Mar 2022 – Jan 2024)
- Re-designed our cloud platform service from the ground up and lead development efforts across the stack
- Rebuilt what was originally an internal tool into an off-the-shelf product, enabling clients to build & monitor themselves

Loko AI

Senior Software Engineer (Sep 2020 – June 2021)
- Head of design, implementation, & deployment efforts for all cloud-based ML software and device firmware
- Designed an ultra-low latency streaming engine for our real-time, rules-based anomaly detection platform
- Integrated CNNs designed for anomaly detection on live video streams into our microservice-based backend
- Development of highly secure, FIPS and NIST compliant software for the military and other high-risk environments
- Projects include an autonomous temperature detector for COVID-19 prevention, used primarily for clinic check-ins

Bridge 12 Technologies

Software & Control Systems Engineer (Aug 2017 – June 2020)
- Chiefly responsible for the control systems of assorted radiation technologies in the terahertz products division
- Projects include particle accelerators, gyrotrons, vacuum bakeout systems, and lab instrumentation
- Wrote the control system software & firmware for proton therapy machines used to treat cancer in hospitals

Skills

Languages

JavaScript	14 years	**HTML**	14 years
TypeScript	12 years	**CSS**	14 years
Python	12 years	**SQL**	12 years
C++	10 years	**C**	8 years
Bash	10 years	**Java**	5 years
Go	4 years	**Kotlin**	2 years

Tools	Node.js, Docker, Kubernetes, Figma, Postman, Django, Flask, NextJS, Cypress, Qt, Terraform
Technologies	AI/ML, REST APIs, UI/UX, CI/CD pipelines, SoM/SoC/FPGA, gRPC, MQTT, NoSQL, cryptography
Infrastructure	AWS (including GovCloud), GCP, Microservices, Serverless, Android, iOS, Linux, Windows
Certifications	AWS Certified AI Practitioner, AWS Certified Cloud Practitioner

Notable Projects

CypherMed Cloud Platform (Promenade Software)

I redesigned the CypherMed Cloud platform from an in-house solution into a scalable, off-the-shelf B2B product that simplifies medical device cloud integration and accelerates time-to-market. I continue to lead a multidisciplinary team in developing and expanding the platform, which now includes a powerful REST API, SDKs in multiple languages, a component library, and an admin dashboard. I oversee its highly scalable cloud infrastructure, ensuring compliance and maintaining a strong security posture to meet the rigorous demands of the medical device industry.

Kolosos Platform (Loko AI)

I was one of the original developers for Kolosos (formerly Caretaker), a real-time computer vision platform that takes video input and instantly provides actionable insights as well as IoT automation for risk management, security, safety and defense. Kolosos was designed to be a highly scalable and performant machine learning tool. Incorporated in the efficient microservice design is my custom stream engine, which allows for secure, ultra-low latency processing of video streams. Variants of the system were also developed for GovCloud and local stacks so as to better serve our military customers and others with stringent security requirements.

Proton Therapy System (Bridge12 Technologies)

During my time at Bridge12, I developed the control system for their latest 1.6 MeV Proton Injector, including the firmware for various embedded subsystems as well as the technician's graphical control software. Injector systems like these are crucial components in cutting-edge proton therapy devices used for cancer treatment. The first of these systems incorporating Bridge12 injectors went on to treat patients in Mass General Hospital.

Education

Tufts University

Medford, MA (2018-2020)

M.S. Mechanical Engineering with a focus in Design and Digital Control

The Pennsylvania State University

University Park, PA (2013-2017)

B.S. Physics with a focus in Computational Physics